Paperwren: A Gated Benchmark for Reading Synthetic Handwritten Forms Across Twelve Layout Families

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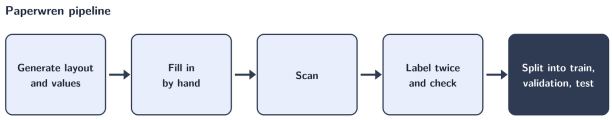


Figure 1: From layout to benchmark.

Table 1: Splits in release v1.0.

Split	Forms
Train	8,640
Validation	1,200
Test	2,560
Total	12,400

Abstract

Paperwren is a benchmark for reading handwritten forms. It contains 12,400 synthetic forms in 12 layout families, each filled in by hand by one of 37 paid annotators and labelled field by field. The task is to return every field's value as text. We release the forms under a data-use agreement, a reference model, and a leaderboard on a held-out test split. The best model we tested reaches 84.6 field-level F1, and most errors come from crossed-out entries and dense tables.

1 Introduction

Forms are still filled in by hand in clinics, libraries and field stations, and reading them is hard for machines: values are crossed out, squeezed into boxes, or written across lines. Paperwren gives a controlled test of this. The layouts and values are synthetic, so nothing private is written on the forms, but the handwriting is real.

2 The dataset

Paperwren v1.0 has 12,400 forms in 12 layout families (Table 1). 37 paid annotators in 3 countries filled them in by hand; every form was labelled field by field and checked by a second annotator.

Table 2: Field-level F1 (%) on the v1.0 test split.

Model	F1
Paperwren-Ref (ours)	84.6
OCR + rules baseline	71.9



Figure 2: Four sample forms.

3 Benchmark

Models return every field's value as text. We score field-level F1 on the held-out test split; test labels are not released. Table 2 gives the results at submission time.

4 Access and ethics

The forms are shared on request under a data-use agreement that forbids identifying annotators from their handwriting. Annotators were paid and gave consent for research use of their handwriting.

5 Limitations

- **English only.** Every form is in English; scores do not transfer to other languages or scripts.
- **Synthetic layouts.** The twelve layout families are designed, not collected; real forms are messier.
- **Small annotator pool.** 37 annotators from 3 countries wrote all forms, so handwriting styles are limited.